Vansh Mishra

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Education

Vellore Institute of Technology Vellore, Tamil Nadu
B.Tech in Computer Science — CGPA: 9.35 (5th Semester) *July 2024 – May 2028*

Nurture International School Kanpur, Uttar Pradesh
Class XII CBSE Board: 94.2% — Class X CBSE Board: 96.5%

Experience

Summer Research Intern – Kotak School of Sustainability Kanpur, Uttar Pradesh
IIT Kanpur, under Prof. Sachchida Nand Tripathi, Dean *15 May 2026 – 15 June 2026*

- Analyzed Scanning Mobility Particle Sizer (SMPS) data from 44 CPCB (Central Pollution Control Board) monitoring sensors across the Delhi region for a project comparing PMF and NMF source apportionment techniques
- Cleaned and processed 14 months of air quality data (April 2024 – June 2025) and built regression models relating PM2.5 concentration to wind speed, wind direction, and temperature
- Applied Positive Matrix Factorization (PMF) and Non-negative Matrix Factorization (NMF) to characterize outdoor aerosol sources from ambient air quality measurements, and generated PM2.5 hotspot maps across Delhi
- Built polar plots from NIMR and ANVT station datasets to visualize pollutant concentration patterns relative to wind direction and speed, contributing findings to ongoing air quality research led by Prof. Sachchida Nand Tripathi at IIT Kanpur

Projects

Void – Modular PDF Processing Backend | *Python, Flask, PyMuPDF, PyPDF2, SQLite* GitHub

- Built a 5-endpoint Flask REST API (merge, split, compress, download, job status) using a modular, service-oriented architecture with factory-pattern extensibility
- Designed a 5-layer validation system enforcing file-type checks, a 100MB size limit, and detection of corrupted, encrypted, and invalid-split-point PDFs
- Engineered a job-based processing pipeline with unique job IDs, status tracking, and download endpoints, persisting job metadata via SQLite
- Implemented centralized logging and automated cleanup utilities; architected the system for extensibility toward DOCX/image conversion, Redis + Celery async jobs, and AWS S3 storage

Multi-Level Cache Performance Analyzer | *Python, Matplotlib, NumPy* GitHub

- Simulated a full L1/L2/L3 cache hierarchy (32KB → 128KB → 512KB) with LRU eviction and a Markov-based AI prefetcher that learns memory access patterns to predict and pre-load future addresses
- Reduced Average Memory Access Time (AMAT) from ∼8.7 to ∼5.9 cycles (∼32% improvement) on sequential traces, while maintaining graceful degradation on random-access patterns
- Built a What-If Cost Analysis module estimating silicon area (mm^2) and power consumption (mW) across varying cache configurations
- Benchmarked sequential, stride, loop, and random access patterns, visualizing AMAT and hit-rate trends via an interactive matplotlib dashboard

Asclepius FastAPI – Medical Symptom Triage API | *Python, FastAPI, Pydantic v2* GitHub

- Built a 6-endpoint RESTful triage API (diagnosis, history retrieval/clearing, health check, echo) using FastAPI and Pydantic v2 for schema validation
- Implemented custom field validators enforcing symptom length, severity scale (1–10), age range (1–120), and duration constraints, alongside middleware and structured exception handling
- Integrated Swagger/OpenAPI documentation for all endpoints and maintained a clean Git history through disciplined rebasing workflows

Technical Skills

Languages: Python, C/C++, Java, R, MATLAB, HTML/CSS
Frameworks: FastAPI, Flask, Streamlit, Vite
Developer Tools: Git, Jupyter Notebook, VS Code, Visual Studio, PyCharm, SQLite
Libraries: pandas, NumPy, scikit-learn, Matplotlib, Pydantic, pytest, seaborn
Concepts: REST API Design, LRU Caching, AI Prefetching, Data Validation, Unit Testing, Random Forest, Linear & Logistic Regression, Decision Trees, Clustering, PMF & NMF Source Apportionment
Problem Solving: Solved 270+ DSA problems on LeetCode (100 Days Badge 2025)